

			10.5 3	A1
6	(a)		A field of force (around a mass) is a <u>region of space</u> that can be <u>mapped with lines of gravitational force (or with lines of gravitational potential)</u> (Answer must be contextualised to “gravitational force” for the first point.) Gravitational field strength at a point is defined as the <u>gravitational force per unit mass</u> acting on a small mass placed at that point.	B1 B1
	(b)		Gravitational force provides the centripetal force for circular motion to take place. By Newton's 2nd Law, $F = ma$ $GMm/r^2 = mr\omega^2$ $GM/r^3 = (2\pi/T)^2$ $T^2 = 4\pi^2 r^3 / GM$	B1 B1 B1
	(c)		$T_G = 7.16$, $\lg T_G = 0.85$ From graph, $\lg (r_G) = 9.05$, $r_G = 1.12 \times 10^9 \text{ m} \sim 1 \times 10^9 \text{ m}$ (estimate)	M1 A1
	(d)		$T_T = (16.2 / 24) \text{ days}$, $\lg T_T = -0.17$	B1

			From graph, $\lg(r_G) = 8.35$, $r_G = 2.24 \times 10^8 \text{ m}$ r_G calculated is the distance of the Thebe to the centre of Jupiter. We can only decide on the accuracy of the statement if the radius of Jupiter is known.	B1
	(e)	(i)	$W = m.\Delta V = m.(\phi_{\text{final}} - \phi_{\text{initial}}) = 53.2(-42.5 - (-47.6))$ $= 271.32 \text{ MJ} = 271.32 \times 10^6 \text{ J} \sim 2.7(1) \times 10^8 \text{ J}$	M1 A1
		(ii)	$\phi = -GM/r$ $\phi.r = \text{constant}$ $42.5(9.38) = 398.65 \sim 400$ $47.6(8.38) = 398.89 \sim 400$ $54.1(7.38) = 399.26 \sim 400$ Need to show constant value, to 1 s.f., for all 3 orbits (shown)	M1
7	(a)		When there is no current through the 12 V battery, the potential	